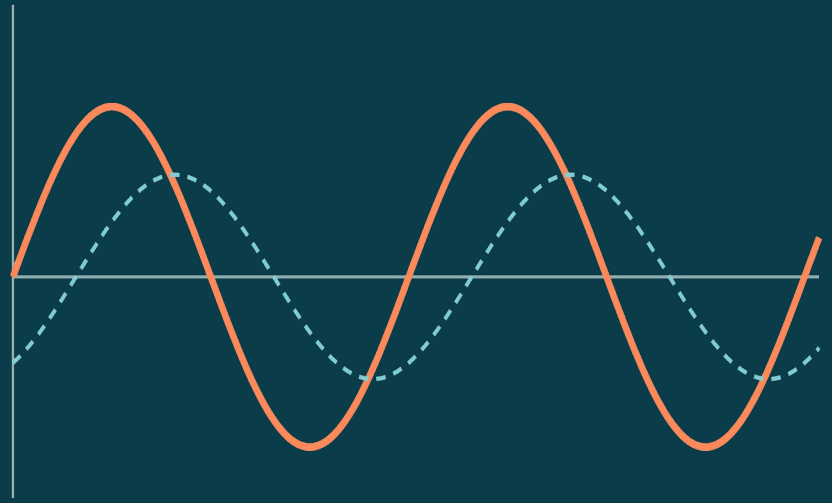
Integrator, Differentiator and Comparator

Basic Electrical and Electronics Engineering

Unit 5: Op-Amp and Linear IC Applications

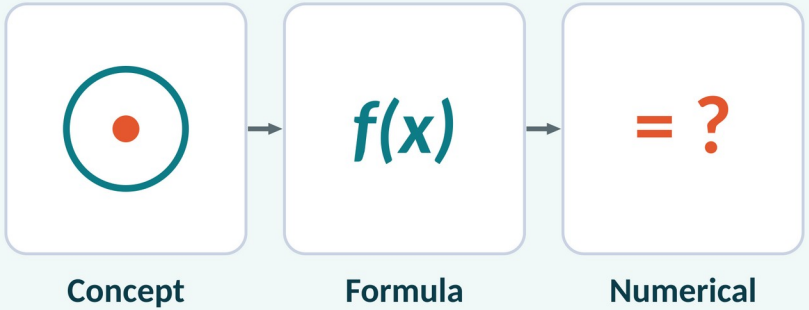
Topic: Op-amp integrator, differentiator and comparator

Prof. Mitesh Astik



Learning Outcomes

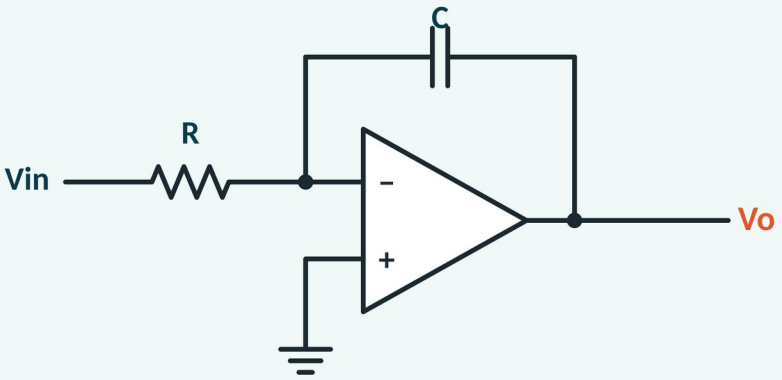
- Derive the output of the integrator and differentiator.
- Predict output waveforms for square and sine inputs.
- Explain the comparator and level detection.
- Calculate output voltages and timing.



Linked to CO5
Describe the working principles of op-amp and signal-processing circuits.

Recap: What You Already Know

- Virtual ground at the inverting input.
- No current enters the op-amp.
- Capacitor: i = C·dv/dt; v = (1/C)∫i dt.
- Inverting amplifier: Vo = −(Zf/Z1)·Vin.



Vo = −(1/RC)·∫Vin dt

virtual ground at (−): current Vin/R charges C

Generalised inverting amplifier: Vo = −(Zf/Z1)·Vin;   capacitor impedance 1/(sC)

Motivation: Calculus with Circuits

- Analog computers solve differential equations.
- Integrators in ADCs and PID controllers.
- Differentiators detect edges and rate of change.
- Comparators turn analog levels into logic.

Circuit	Input → output	Formula
Integrator	square → triangle	$-(1/RC)\int V_{in}\,dt$
Differentiator	triangle → square	$-RC\cdot dV_{in}/dt$
Comparator	sine → square	$\pm V_{sat}$

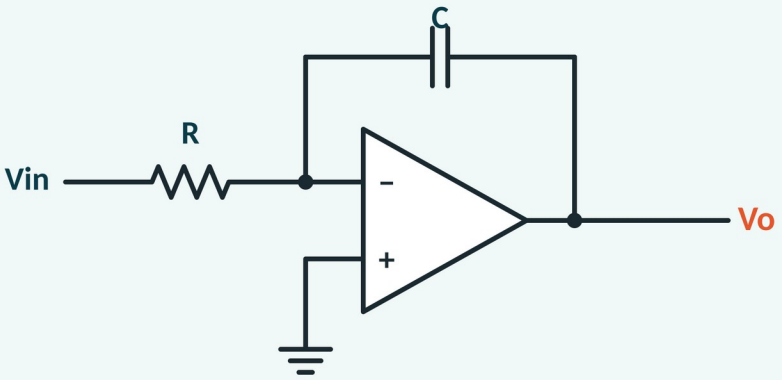
Uses: wave shaping, ADC, filters, level detection

Integrator: low-pass; differentiator: high-pass

Three basic functions: ∫, d/dt and decision (comparison)

Op-Amp Integrator

- Resistor R at the input; capacitor C in feedback.
- Input current Vin/R charges the capacitor.
- Capacitor voltage is the integral of that current.
- Output is minus the integral of the input.
- Gain falls with frequency: low-pass.



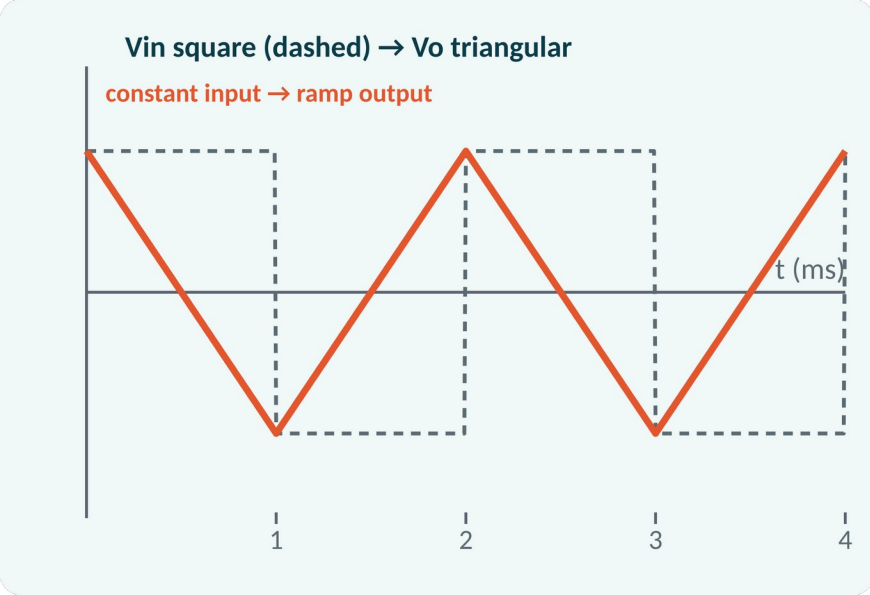
Vo = −(1/RC)·∫Vin dt

virtual ground at (−): current Vin/R charges C

Vo(t) = −(1/RC)·∫Vin dt + Vo(0); RC time constant (s)
In s-domain: Vo/Vin = −1/(sRC)

Integrator: Waveforms and Limitations

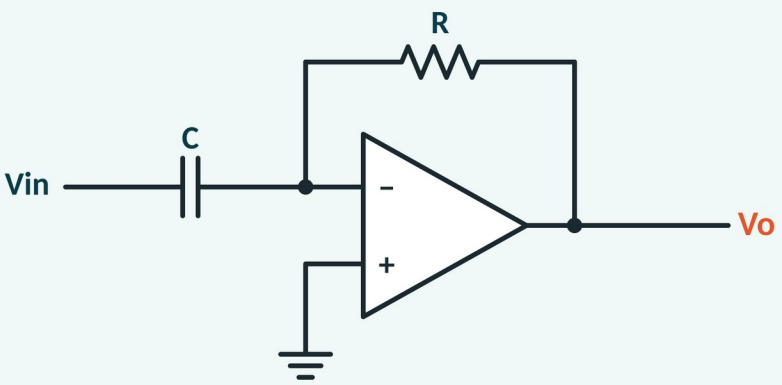
- DC input → linear ramp.
- Square-wave input → triangular output.
- Sine input → cosine (90° shifted).
- Offset causes drift until saturation.
- Fix: resistor Rf across C limits DC gain.



Square ±V, half-period T/2: output swing = V·T/(2RC) (peak-to-peak)
Practical integrator: Rf ≈ 10 R in parallel with C

Op-Amp Differentiator

- Capacitor C at the input; resistor R in feedback.
- Capacitor current = C·dVin/dt.
- That current flows through R.
- Output is minus RC times the rate of change.
- Gain rises with frequency: high-pass.



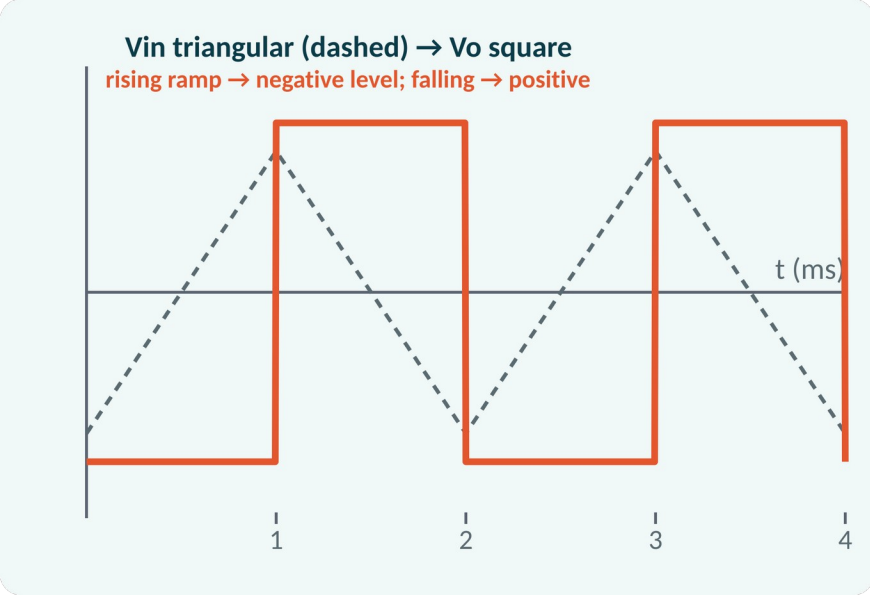
Vo = −R·C·dVin/dt

output ∝ rate of change of the input

Vo(t) = −R·C·dVin/dt; in s-domain: Vo/Vin = −sRC

Differentiator: Waveforms and Limitations

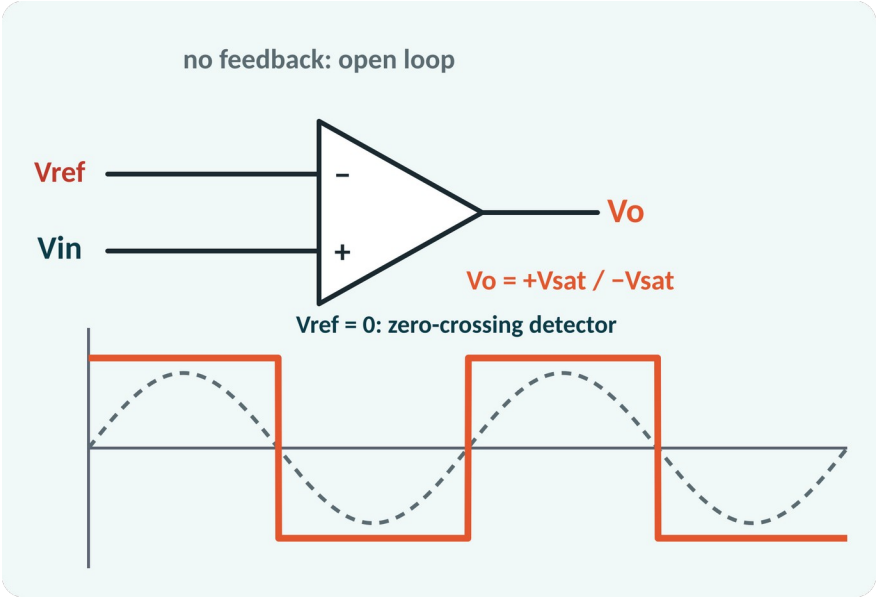
- Triangular input → square output.
- Sine input → cosine output (amplitude × RCω).
- Amplifies high-frequency noise strongly.
- Can become unstable.
- Practical: series Rs with C and Cf across R.



Vin = Vm sin ωt → Vo = −RC·ωVm cos ωt
Practical differentiator: add Rs (series with C) and Cf (across Rf)

Op-Amp Comparator

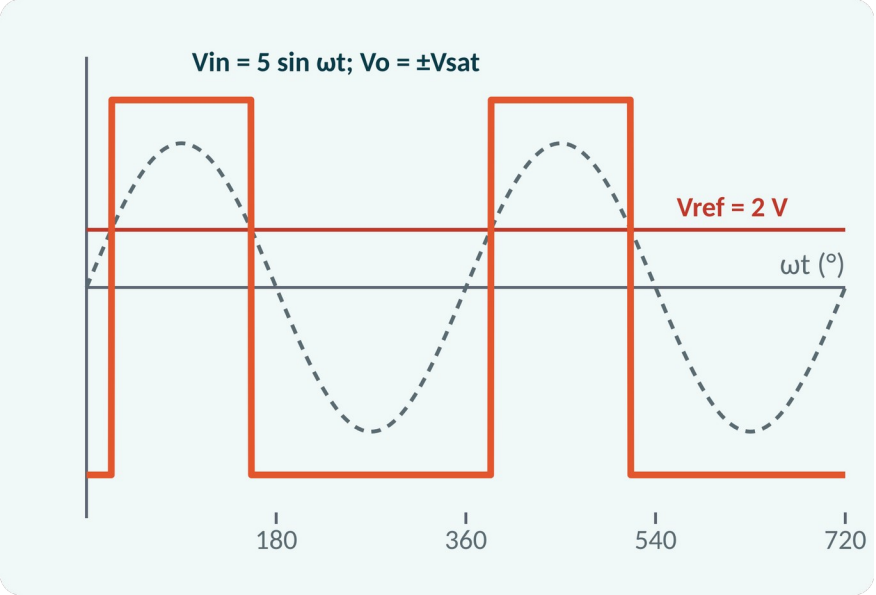
- Op-amp used without feedback (open loop).
- Compares Vin with a reference Vref.
- Vin > Vref: output +Vsat.
- Vin < Vref: output −Vsat.
- Zero-crossing detector: Vref = 0.



Vo = +Vsat if V+ > V−; Vo = −Vsat if V+ < V−; Vsat ≈ VCC − 1.5 V

Level Detector and Hysteresis

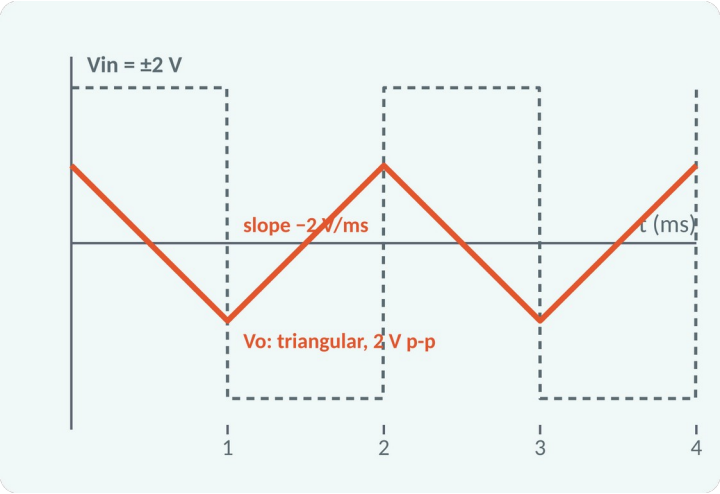
- Non-zero Vref sets the trigger level.
- Output is high only while Vin > Vref.
- Pulse width depends on Vref.
- Noise near the threshold causes chatter.
- Positive feedback (Schmitt trigger) adds hysteresis.



Switching angle: $\theta = \sin^{-1}(V_{ref}/V_m)$; high for $180° - 2\theta$ each cycle

Solved Example 1: Integrator

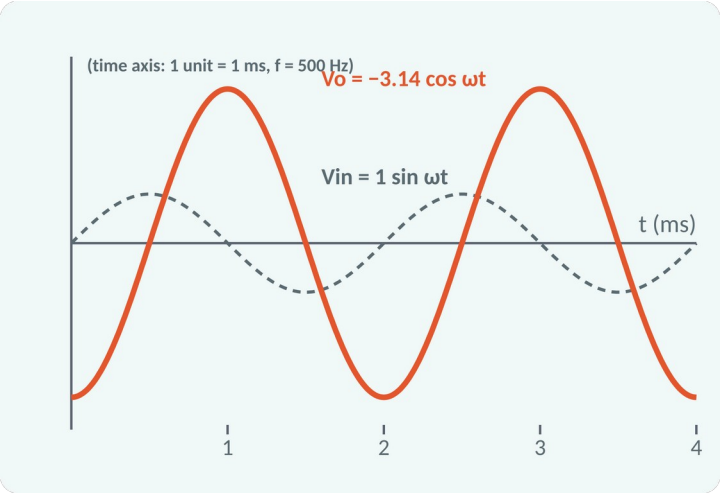
- R = 10 kΩ, C = 0.1 µF. Input: ±2 V square wave, frequency 500 Hz (half-period 1 ms). Find the output.
- 1. RC = 10 kΩ × 0.1 µF = 1 ms.
- 2. Slope = −Vin/RC = −2 V/1 ms = −2 V/ms (for +2 V).
- 3. In 1 ms the output falls by 2 V; with −2 V it rises by 2 V.
- 4. Output: triangular wave, 2 V peak-to-peak.



Answer: triangular output, slope ∓2 V/ms, 2 V peak-to-peak (±1 V)

Solved Example 2: Differentiator

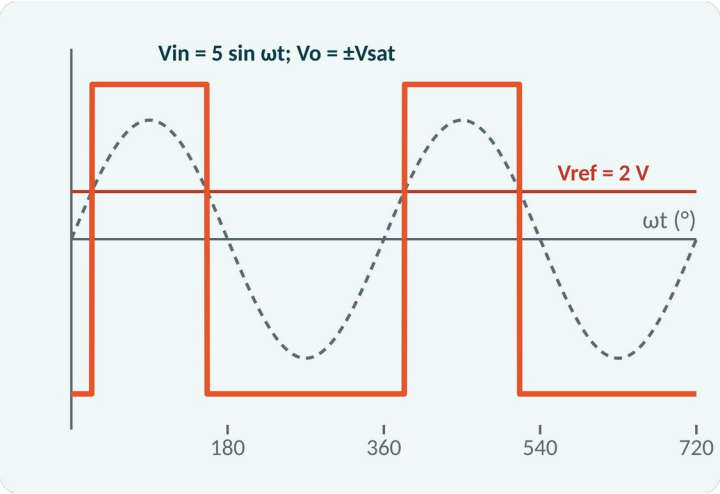
- R = 10 kΩ, C = 0.1 µF, Vin = 1·sin(2π × 500t) V. Find Vo.
- 1. RC = 1 ms; ω = 2π × 500 = 3142 rad/s.
- 2. dVin/dt = 1 × ω·cos ωt = 3142 cos ωt V/s.
- 3. Vo = −RC·dVin/dt = -10^{-3} × 3142 cos ωt.
- 4. Vo = −3.14 cos(2π × 500t) V.



Answer: Vo = −3.14 cos ωt V (amplitude 3.14 V, 90° shifted, inverted)

Solved Example 3: Level Detector

- Vin = 5 sin ωt at the (+) input, Vref = +2 V at the (−) input, Vsat = ±13 V. Find the switching angles and the duty cycle.
- 1. Output high when 5 sin ωt > 2.
- 2. sin θ = 2/5 = 0.4 → θ1 = 23.6°.
- 3. θ2 = 180° − 23.6° = 156.4°.
- 4. High for 156.4° − 23.6° = 132.8° per cycle.
- 5. Duty = 132.8/360 = 36.9 %.



Answer: Vo = +13 V for 23.6° < ωt < 156.4°, else −13 V; duty 36.9 %

Common Mistakes and Exam Tips

Common mistakes

- Forgetting the minus sign in the integrator and differentiator.
- Using RC in the wrong units (μF, kΩ).
- Expecting a stable differentiator at high frequency.
- Ignoring op-amp saturation in the integrator.
- Using feedback in a comparator.

Exam tips

- $V_o = -(1/RC)\int V_{in}\,dt$; $V_o = -RC \cdot dV_{in}/dt$.
- Write RC in seconds first.
- Check that $|V_o| < V_{sat}$.
- Comparator: no feedback; output ±Vsat.

Summary

- Integrator: Vo = −(1/RC)∫Vin dt (low-pass).
- Differentiator: Vo = −RC dVin/dt (high-pass).
- Square → triangle (integrator); triangle → square (differentiator).
- Comparator: Vo = ±Vsat depending on Vin ≷ Vref.
- Practical versions need Rf / Rs and Cf.

Circuit	Input → output	Formula
Integrator	square → triangle	−(1/RC)∫Vin dt
Differentiator	triangle → square	−RC·dVin/dt
Comparator	sine → square	±Vsat

Uses: wave shaping, ADC, filters, level detection

Integrator: low-pass; differentiator: high-pass

∫: slope = −Vin/RC; d/dt: amplitude = RC·ω·Vm

Quick Quiz

Q1. Integrator with square-wave input gives:
(a) square (b) triangular (c) sine (d) DC

Ans: (b)

Q2. In a differentiator the capacitor is:
(a) in feedback (b) at the input (c) to ground (d) absent

Ans: (b)

Q3. RC = 1 ms, Vin = 5 V DC: integrator slope =
(a) −5 V/ms (b) +5 V/ms (c) −5 mV/ms (d) 0

Ans: (a)

Q4. A comparator with Vin > Vref (Vin at +) gives:
(a) 0 V (b) −Vsat (c) +Vsat (d) Vref

Ans: (c)

Q5. Differentiator output for sin input is:
(a) sine (b) cosine (c) DC (d) triangular

Ans: (b)

References

1. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3, 4, 5 textbook)

2. **David A. Bell, Operational Amplifier and Linear ICs**
   3rd Ed., Oxford University Press, 2011 (reference, Unit 5)

Next lecture: 555 timer and 78xx voltage regulators